THE FOURIER ENTROPY–INFLUENCE INEQUALITY FOR MONOTONE DNF FORMULAS OF BOUNDED WIDTH

YITZCHAK SHMALO

Abstract. The Fourier entropy–influence conjecture of Friedgut and Kalai asserts that the spectral entropy of a Boolean function is at most a universal constant times its total influence. It is open even for monotone functions. We prove it with constant $w + 8.1$ for every monotone DNF formula and every monotone CNF formula of width w, with no restriction on the number of terms or on the number of times a variable occurs; for width two, that is for the indicator that a random set of vertices of a graph contains an edge, the constant is 9.41. More generally the bound holds for every class of monotone functions that is closed under restrictions and whose nonconstant members have a minterm or a maxterm of size at most w. The proof works with the uniform measure on the set of zeros of the function, on which every coordinate is a fair coin unless it is blocked. This gives a bound on the entropy lost when one coordinate is restricted, in terms of its influence relative to the variance. It also bounds the entropy by the influences measured relative to the variance, and it reduces the conjecture for monotone functions to monotone functions all of whose influences are small compared with the variance. Together with the influential-variable theorem of O'Donnell, Saks, Schramm and Servedio it gives the constant $\log_2 \log_2 s + 7.85$ for monotone functions computed by decision trees with s leaves, and $\log_2 m + 8.85$ for monotone DNF formulas with m terms. The same classes satisfy the stronger inequality of O'Donnell and Tan, also under product measures with biases bounded away from 0 and 1. So the conjecture also holds for read-once formulas whose leaves are monotone DNFs or CNFs of bounded width and whose gates are AND, OR, XOR or threshold gates of any fan-in, or bounded-width monotone DNFs whose inputs have biases bounded away from 0 and 1.

1. Introduction

For $f\colon \{0,1\}^n \to \{0,1\}$ let $F = 1 - 2f$, $\chi_T(x) = (-1)^{\sum_{i\in T} x_i}$, $\widehat{F}(T) = \mathbb{E}[F\chi_T]$ with x uniform, and $P(T) = \widehat{F}(T)^2$. By Parseval P is a probability distribution on the subsets of $[n]$; a random set S with law P is the *spectral sample* of f. The *spectral entropy* and the *total influence* are

$$H(f) = \sum_T P(T) \log_2 \frac{1}{P(T)}, \qquad I(f) = \sum_T |T|\, P(T) = \sum_v \mathrm{Inf}_v(f),$$

where $\mathrm{Inf}_v(f) = \mathbb{P}[f(x^{v\leftarrow 1}) \neq f(x^{v\leftarrow 0})] = \sum_{T\ni v} P(T)$. Friedgut and Kalai [4] conjectured that $H(f) \le C\, I(f)$ for a universal constant C. The conjecture is known for symmetric functions and read-once decision trees [12], for read-once formulas [11], for random linear threshold functions [2], for functions of very small influence or very large entropy [13], for δ-tribes and functions with separation properties [5], and for read-k DNF formulas and regular functions with constants $O(\log k)$ and 15.1 [14]. The best general bounds lose a factor: $H(f) = O(I(f) + \sum_A P(A)|A| \log |A|)$ [8], and $H(f) \le \sum_v h(\mathrm{Inf}_v) = O(I(f) + \sum_v \mathrm{Inf}_v \log(1/\mathrm{Inf}_v))$, with h the binary entropy, which follows from the subadditivity of entropy [9] and is also proved in [7].

The conjecture is open for monotone functions. There it would give total influence $\Omega(\log^2 n)$ for every monotone graph property on n vertices with constant variance [4], where the best bound known is $\Omega(\log^2 n/(\log\log n)^2)$ [8]. The monotone case of the proofs above rests on an

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order of the coordinates by importance (regular functions), on disjointness (read-once formulas) or on bounded overlap of terms (read-k DNFs). The results here use none of these.

For f monotone write $q = \mathbb{P}[f = 0]$ and $m = \min(q, 1-q)$; then $\mathrm{Var}(F) = 4q(1-q)$ lies between $2m$ and $4m$. A *minterm* of f is a minimal set M with $f(\mathbf{1}_M) = 1$, a *maxterm* a minimal set M with $f(\mathbf{1}_{[n]\setminus M}) = 0$. Let

$$\Psi(r) = \log_2\bigl(e\,(r + 2\sqrt{r} + 2)\bigr) \quad (r \geq 0), \qquad c_w = \frac{2}{\ln 2} + \frac{2e^{-1/2}}{\ln 2} + \Psi(2^{w+1} - 4).$$

Numerically $2/\ln 2 + 2e^{-1/2}/\ln 2 = 4.6355$, $c_1 = 7.078$, $c_2 = 9.400$, $c_3 = 10.466$, $c_4 = 11.421$, and $c_w \leq w + 8.1$ for every $w \geq 1$.

Theorem 1.1. *Let $\mathcal{C}$ be a class of monotone Boolean functions that is closed under restrictions of variables to constants and such that every nonconstant $f \in \mathcal{C}$ has a minterm or a maxterm with at most w elements. Then $H(f) \leq c_w\, I(f)$ for every $f \in \mathcal{C}$.*

Corollary 1.2. *Every monotone DNF formula whose terms have at most w variables, and every monotone CNF formula whose clauses have at most w variables, satisfies $H(f) \leq (w+8.1)\, I(f)$. For $w = 2$ the constant is 9.41: for every graph G on the vertex set $[n]$ and every $U \subseteq [n]$, the function*

$$f(x) = \bigvee_{u \in U} x_u \;\vee \bigvee_{uw \in E(G)} (x_u \wedge x_w)$$

satisfies $H(f) \leq 9.41\, I(f)$, and so does the indicator that the set of ones of x is a vertex cover of G.

The class of DNFs of width w is closed under restrictions: setting $x_v = 0$ deletes the terms containing v and setting $x_v = 1$ removes v from them. Every term is an implicant, so a nonconstant such f has a minterm of size at most w; the CNF case is the dual. For the graph formula, $1 - f$ is the indicator that the set of ones is an independent set of G avoiding U. For this class the read-k theorem of [14], with k the maximum degree of G, gives $(30 + 5.6 \log_2 k)\, I(f)$. Another member of the class of Theorem 1.1: for a fixed graph K with e edges, the property that a graph on n vertices contains a copy of K, as a function of the $\binom{n}{2}$ edge indicators, is a monotone DNF of width e, so its spectral entropy is at most $(e + 8.1)$ times its total influence, uniformly in n.

Theorem 1.1 follows from a bound on the restriction step at a single coordinate. For a coordinate v let f_0, f_1 be the restrictions $x_v = 0, 1$ and

$$\Delta_v(f) = H(f) - \tfrac{1}{2}\bigl(H(f_0) + H(f_1)\bigr).$$

Since $I(f) = \mathrm{Inf}_v(f) + \frac{1}{2}(I(f_0) + I(f_1))$, a class of functions closed under restrictions in which every nonconstant member has a coordinate with $\Delta_v \leq K \mathrm{Inf}_v$ satisfies $H \leq KI$, by induction on the number of variables; this is the induction used by González, MacManus and Pereyra [5].

Theorem 1.3. *Let f be monotone and let v be a coordinate with $\mathrm{Inf}_v(f) > 0$. Then*

$$\Delta_v(f) \leq \mathrm{Inf}_v(f)\Bigl(\frac{2}{\ln 2} + \frac{2e^{-1/2}}{\ln 2} + \Psi\Bigl(\frac{4m}{\mathrm{Inf}_v(f)} - 2\Bigr)\Bigr).$$

For an arbitrary Boolean function the step is at most $(2/\ln 2 + \log_2(e/\mathrm{Inf}_v))\mathrm{Inf}_v$ when $\mathrm{Inf}_v \leq \frac{1}{2}$; summed along a telescoping it gives the bound of [9] up to the constant. For monotone functions the logarithm is of the influence relative to the variance. It cannot be removed altogether (Remark 7.1). Summing the step along the telescoping of [7] gives the following.

Corollary 1.4. *For every monotone f,*

$$H(f) \le \sum_v \mathrm{Inf}_v(f)\Big(4.6355 + \Psi\Big(\frac{4m}{\mathrm{Inf}_v(f)} - 2\Big)\Big) \le 7.08\, I(f) + \sum_v \mathrm{Inf}_v(f) \log_2 \frac{4m}{\mathrm{Inf}_v(f)}.$$

For balanced functions this is no better than the bound $H(f) \le \sum_v h(\mathrm{Inf}_v)$. For biased functions it is stronger: for the conjunction of n variables every influence equals $2m$ and the right side is $8.08\, I$.

Corollary 1.5. *Let $0 < \tau \le 2$ and $K > 0$. Suppose every monotone g with $\mathrm{Inf}_v(g) < \tau\, m(g)$ for all v has a coordinate with $\Delta_v(g) \le K\mathrm{Inf}_v(g)$. Then every monotone f satisfies $H(f) \le \max(K, 4.6355 + \Psi(4/\tau - 2))\, I(f)$.*

So the conjecture for monotone functions reduces to monotone functions whose influences are all small compared with their variance. Balanced tribes, majority and monotone graph properties with constant variance are of this kind; Section 7 explains why the argument gives nothing more there.

In Theorem 1.1 the large influence comes from a short minterm. Decision trees supply another: by the theorem of O'Donnell, Saks, Schramm and Servedio [10], a function computed by a decision tree has a coordinate whose influence is at least the variance divided by the expected depth of the tree. For a Boolean function g let $\bar{d}(g)$ be the least expected depth, on a uniformly random input, of a decision tree computing g, and let $L(f)$ be the largest $\bar{d}(g)$ over the restrictions g of f. A tree may be assumed to read no coordinate twice on a path; then a leaf at depth k is reached with probability 2^{-k}, and the expected depth is the entropy of the leaf reached. Restricting a tree does not add leaves, so $L(f) \le \log_2 s$ if f is computed by a tree with s leaves. Also $L(f)$ is at most the depth of any tree for f and, averaging over the random choices, at most the least worst-case expected number of queries of a randomized decision tree for f.

Theorem 1.6. *Every nonconstant monotone f satisfies*

$$H(f) \le \big(4.6355 + \Psi(2L(f) - 2)\big)\, I(f) \le \big(\log_2 L(f) + 7.85\big)\, I(f).$$

In particular, a monotone function computed by a decision tree with $s \ge 2$ leaves satisfies $H(f) \le (\log_2 \log_2 s + 7.85)\, I(f)$.

For Boolean functions with $I(f) \ge 1$ computed by a decision tree of expected depth d, Wan, Wright and Wu [15] proved $H(f) \le 12d\, I(f)$. For monotone functions Theorem 1.6 replaces the factor d by its logarithm, at the price of a maximum over restrictions. Write $s(f)$ for the sensitivity of f.

Corollary 1.7. *Let f be a nonconstant monotone function.*

(a) *If f is a DNF formula with m terms or a CNF formula with m clauses, then $H(f) \le (\log_2 m + 8.85)\, I(f)$.*
(b) *If the minterms of f have at most w_1 elements and its maxterms at most w_0 elements, then $H(f) \le (\log_2(w_0 w_1) + 7.85)\, I(f)$.*
(c) *$H(f) \le (2 \log_2 s(f) + 7.85)\, I(f)$.*
(d) *If f has Fourier degree k, then $H(f) \le (4 \log_2 k + 9.85)\, I(f)$.*

For DNF formulas with m terms, monotone or not, the read-k theorem of [14] gives $(30 + 5.6 \log_2 m)\, I(f)$. Part (b) is logarithmic in the widths, where Theorem 1.1 is linear in the smaller one.

O'Donnell and Tan [11] introduced a stronger inequality that is preserved under disjoint composition. With $H^+(f) = \sum_{A \ne \emptyset} P(A) \log_2(1/P(A))$ and $\mathrm{Var}(f) = 1 - P(\emptyset)$ it reads

$$H^+(f) \le C\,\big(I(f) - \mathrm{Var}(f)\big), \tag{1}$$

and it implies $H(f) \le \max(C, 1/\ln 2)\, I(f)$ [11, Proposition 2.1].

Theorem 1.8. *Every f in a class $\mathcal{C}$ as in Theorem 1.1 satisfies (1) with*

$$C = C_w^+ = \max\Big(8.80,\ 2\Big(1.0615 + \frac{2}{\ln 2} + \Psi(2^{w+1} - 4)\Big)\Big).$$

Numerically $C_1^+ = 12.78$, $C_2^+ = 17.43$, $C_3^+ = 19.56$, and $C_w^+ \le 2w + 14.8$ for every $w \ge 1$. Every nonconstant monotone f satisfies (1) with $C = \max\big(8.80,\ 2(1.0615 + 2/\ln 2 + \Psi(2L(f) - 2))\big) \le 2\log_2 L(f) + 14.33$.

By the composition theorem of [11], functions satisfying (1) can be combined along a read-once formula whose gates satisfy the biased form of (1) at the means of their inputs. Under every product measure this holds for AND, OR and XOR gates of any fan-in with constant $2 + 4\log_2 e < 7.78$, for threshold gates $\mathbf{1}[y_1 + \cdots + y_r \ge t]$ of any fan-in with constant 760 [14], and for gates of arity at most a with a constant $2^{O(a)}$ [11].

Corollary 1.9. *Let F be a read-once formula and $f = F(f_1, \dots, f_s)$, where $f_1, \dots, f_s$ are monotone DNF or CNF formulas of width at most w on pairwise disjoint sets of variables. Suppose every gate of F is an AND, OR, XOR or threshold gate of any fan-in, or a monotone DNF or CNF formula of width at most w whose inputs are balanced, that is, equal to 1 with probability $\frac{1}{2}$. Then $H(f) \le \max(C_w^+, 760)\, I(f)$, and $H(f) \le C_w^+\, I(f)$ if there are no threshold gates.*

For instance, the conjunction of any number of monotone 2-DNFs on pairwise disjoint sets of variables is a monotone DNF of unbounded width, and it satisfies $H \le 17.43\, I$; so does $G(\mathrm{Maj}_3(x^1), \dots, \mathrm{Maj}_3(x^m))$ for every monotone 2-DNF G and disjoint blocks $x^1, \dots, x^m$, since the majority of three unbiased bits is unbiased. By the last statement of Theorem 1.8, the leaves f_j in Corollary 1.9 may also be monotone functions with $L(f_j) \le L$, with C_w^+ replaced by $\max(C_w^+, 2\log_2 L + 14.33)$.

The argument also works under product measures whose biases stay away from 0 and 1. For a product measure μ let $P_\mu(A)$ be the squared coefficients of F in the orthonormal basis of $L^2(\mu)$, I_μ and Var_μ the corresponding total influence and variance, and $\sigma_A^2 = \prod_{i \in A} 4\mu(x_i = 1)\mu(x_i = 0)$ (Section 6).

Theorem 1.10. *Let $0 < \eta \le \frac{1}{2}$, let μ be a product measure with $\mu(x_i = 1) \in [\eta, 1 - \eta]$ for every i, and let $\mathcal{C}$ be as in Theorem 1.1. Then every $f \in \mathcal{C}$ satisfies*

$$\sum_{A \ne \emptyset} P_\mu(A) \log_2 \frac{\sigma_A^2}{P_\mu(A)} \le \big(14.24 + 2w \log_2(1/\eta)\big)\big(I_\mu(f) - \mathrm{Var}_\mu(f)\big).$$

So in Corollary 1.9 a gate that is a monotone DNF or CNF formula of width at most w may have inputs whose biases lie in $[\eta, 1 - \eta]$, with C_w^+ replaced by $\max(C_w^+, 14.24 + 2w\log_2(1/\eta))$.

2. The restriction step

2.1. **Erasure and Jensen–Shannon parts.** Fix v. For $T \subseteq [n] \setminus \{v\}$ the sets T and $T \cup \{v\}$ carry the weights $P(T)$ and $P(T \cup v)$, and the restrictions have $\widehat{F_b}(T) = \widehat{F}(T) \pm \widehat{F}(T \cup v)$. Let $\phi(t) = t\log_2(1/t)$ and $\mathcal{E}(X, Y) = \phi(X) + \phi(Y) - \phi(X + Y) = (X + Y)\, h\big(Y/(X + Y)\big)$, where h is the binary entropy. The law of $S \setminus \{v\}$ is the average of the spectral distributions of F_0 and F_1, so

$$\Delta_v = B_v + \mathrm{JS}_v, \qquad B_v = \sum_{T \not\ni v} \mathcal{E}\big(P(T), P(T \cup v)\big) = H(\mathbf{1}[v \in S] \mid S \setminus \{v\}),$$

and JS_v is the Jensen–Shannon divergence of the two spectral distributions.

Lemma 2.1. *For every Boolean function,* $\mathrm{JS}_v \le \frac{2}{\ln 2}\sum_{T\not\ni v}\min\big(P(T),P(T\cup v)\big) \le \frac{2}{\ln 2}\mathrm{Inf}_v$.

Proof. Put $u=\widehat{F}(T)$ and $w=\widehat{F}(T\cup v)$. The pair contributes $J=\phi(u^2+w^2)-\frac12\phi((u+w)^2)-\frac12\phi((u-w)^2)$, which is $1/\ln 2$ times the entropy $\mathrm{Ent}(g^2)=\mathbb{E}g^2\ln g^2-\mathbb{E}g^2\ln\mathbb{E}g^2$ of the function g on the two-point space with the uniform measure and values $|u+w|$, $|u-w|$. By the logarithmic Sobolev inequality of the two-point space [6], $\mathrm{Ent}(g^2)\le\frac12(|u+w|-|u-w|)^2=2\min(u^2,w^2)$. □

2.2. **The down-set identity.** Let f be monotone, $D=\{x: f(x)=0\}$, $q=|D|2^{-n}>0$, and let μ be the uniform measure on D, with $\tilde\mu(T)=\mathbb{E}_\mu\chi_T$. Since $F=2\cdot\mathbf{1}_D-1$,

$$\widehat{F}(T)=2q\,\tilde\mu(T)\qquad(T\neq\emptyset). \tag{2}$$

Fix v. For $y\in\{0,1\}^{[n]\setminus\{v\}}$ call y *blocked* if $y\in D$ and $y+v\notin D$, and *free* if $y+v\in D$; here $y+v$ is y with the v-th coordinate set to 1. As D is a down-set, the fibre of D over a blocked y is $\{y\}$ and over a free y is $\{y,y+v\}$. For $T\subseteq[n]\setminus\{v\}$ put

$$a(T)=\frac{1}{|D|}\sum_{y\ \text{blocked}}\chi_T(y),\qquad b(T)=\frac{2}{|D|}\sum_{y\ \text{free}}\chi_T(y).$$

Lemma 2.2. *For* $T\subseteq[n]\setminus\{v\}$, $\tilde\mu(T\cup v)=a(T)$ *and* $\tilde\mu(T)=a(T)+b(T)$. *With* $\varepsilon=\mathrm{Inf}_v(f)/(4q)$,

$$q\sum_T a(T)^2=\varepsilon,\qquad q\sum_T b(T)^2=1-2\varepsilon,\qquad 0\le\varepsilon\le\tfrac12.$$

Proof. On a free fibre x_v takes both values with equal weight, so $\chi_{T\cup v}$ averages to 0 there, while on a blocked fibre $x_v=0$; this gives both identities. Let N_B,N_F be the numbers of blocked and free y. Then $|D|=N_B+2N_F$, and $\mathrm{Inf}_v=N_B/2^{n-1}$ because the edges of the cube in direction v on which f changes are exactly the pairs $\{y,y+v\}$ with y blocked. Parseval on $\{0,1\}^{n-1}$ gives $\sum_T a(T)^2=2^{n-1}N_B/|D|^2$ and $\sum_T b(T)^2=2^{n-1}\cdot 4N_F/|D|^2$; multiplying by $q=|D|/2^n$ gives $N_B/(2|D|)=\varepsilon$ and $2N_F/|D|=1-2\varepsilon$. □

Under μ a coordinate is a fair coin on free fibres and 0 on blocked ones, and 2ε is the μ-probability that it is blocked. By (2), $P(T\cup v)=4q^2a(T)^2$ for every T and $P(T)=4q^2(a(T)+b(T))^2$ for T nonempty. Also $\widehat{F}(\{v\})=\mathrm{Inf}_v$ for monotone f, so $P(\{v\})=\mathrm{Inf}_v^2$.

2.3. **The erasure bound.** The function Ψ is increasing and concave, since $r+2\sqrt{r}+2$ is concave and positive. Let $c_0=\max_{0<x\le1}x\log_2(e/x^2)=2e^{-1/2}/\ln 2=1.75008$.

Lemma 2.3. *For monotone* f *and* $\mathrm{Inf}_v(f)>0$, $B_v\le\mathrm{Inf}_v\big(c_0+\Psi(4m/\mathrm{Inf}_v-2)\big)$.

Proof. The dual $f^*(x)=1-f(\mathbf{1}-x)$ is monotone, its coefficients are $-(-1)^{|T|}\widehat{F}(T)$, so it has the same P, and $\mathbb{P}[f^*=0]=1-q$. Hence it suffices to prove the bound with q in place of m; note $\mathrm{Inf}_v\le 2q$. We use $\mathcal{E}(X,Y)\le Y\log_2\big(e(X+Y)/Y\big)$, which follows from $X\log_2(1+Y/X)\le Y\log_2 e$.

For $T=\emptyset$, $P(\{v\})=\mathrm{Inf}_v^2$ and $P(\emptyset)+P(\{v\})\le 1$, so this pair contributes at most $\mathrm{Inf}_v^2\log_2(e/\mathrm{Inf}_v^2)\le c_0\mathrm{Inf}_v$.

Let $T\neq\emptyset$. If $a(T)=0$ the pair contributes 0. Otherwise let $s=b(T)/a(T)$. Since $\mathcal{E}$ is homogeneous of degree one, the pair contributes $4q\cdot q\,a(T)^2\cdot\psi(s)$, where $\psi(s)=\mathcal{E}((1+s)^2,1)\le\log_2\big(e((1+s)^2+1)\big)\le\Psi(s^2)$. The weights $w_T=q\,a(T)^2$ have total $W\le\varepsilon$, and $\sum_T w_T s_T^2\le q\sum_T b(T)^2=1-2\varepsilon$ by Lemma 2.2. By Jensen's inequality for the concave increasing Ψ,

$$\sum_T w_T\Psi(s_T^2)\le W\,\Psi\Big(\frac{1-2\varepsilon}{W}\Big)\le\varepsilon\,\Psi\Big(\frac{1-2\varepsilon}{\varepsilon}\Big),$$

where the last step holds because $W \mapsto W\Psi(c/W)$ is nondecreasing: its derivative is $\Psi(z) - z\Psi'(z)$ at $z = c/W$, which is at least $\Psi(0) > 0$ by concavity. As $(1-2\varepsilon)/\varepsilon = 4q/\mathrm{Inf}_v - 2$ and $4q\varepsilon = \mathrm{Inf}_v$, the pairs with $T \neq \emptyset$ contribute at most $\mathrm{Inf}_v \Psi(4q/\mathrm{Inf}_v - 2)$. □

Theorem 1.3 is Lemma 2.3 together with Lemma 2.1.

3. Short certificates and decision trees

Lemma 3.1. *Let f be monotone with a minterm M of size $k \geq 1$. Then some $v \in M$ has $\mathrm{Inf}_v(f) \geq 4q/(2^{k+1} - 2)$. If f has a maxterm of size k, some v in it has $\mathrm{Inf}_v(f) \geq 4(1-q)/(2^{k+1} - 2)$.*

Proof. The second statement is the first for f^*. If $k = 1$, then $M = \{v\}$ and $f = x_v \vee g$ with g not depending on x_v, so $\mathrm{Inf}_v = \mathbb{P}[g = 0] = 2q$. Let $k \geq 2$, $M = \{v_1, \ldots, v_k\}$, and let δ be the largest of the numbers $\varepsilon_{v_j} = \mathrm{Inf}_{v_j}/(4q)$. Under μ the conditional probability that $x_{v_j} = 1$ given the other coordinates is $\frac{1}{2}$ on free fibres and 0 on blocked ones, and the blocked fibres have probability $2\varepsilon_{v_j}$. Hence $p_j = \mu(x_{v_2} = \cdots = x_{v_j} = 1)$ satisfies $p_1 = 1$ and $p_{j+1} \geq \frac{1}{2}(p_j - 2\delta)$, so that $p_k \geq 2^{1-k} - \delta(2 - 2^{2-k})$. When $x_{v_2} = \cdots = x_{v_k} = 1$ the coordinate v_1 is blocked, since adding it completes M; hence $2\delta \geq 2\varepsilon_{v_1} \geq p_k$, and therefore $\delta \geq 1/(2^{k+1} - 2)$. □

For $k = 2$ the bound reads $\varepsilon \geq \frac{1}{6}$ at one end of every edge; it is attained by a single edge, where both ends are occupied with probability $\frac{1}{3}$ under the uniform measure on independent sets.

Proof of Theorem 1.1. Let $f \in \mathcal{C}$ be nonconstant. By Lemma 3.1 it has a coordinate with $\mathrm{Inf}_v \geq 4m/(2^{w+1} - 2)$, so $4m/\mathrm{Inf}_v - 2 \leq 2^{w+1} - 4$ and Theorem 1.3 gives $\Delta_v \leq c_w \mathrm{Inf}_v$, as Ψ is increasing. The restrictions are in $\mathcal{C}$ and have fewer variables, and constants have $H = I = 0$; induction completes the proof. For the estimate $c_w \leq w + 8.1$ use $\Psi(r) \leq \log_2 e + \log_2(r + 2\sqrt{r} + 2)$ with $r = 2^{w+1} - 4$, which gives $c_w \leq w + 1 + \log_2 e + 4.6355 + \log_2(1 + 2^{(1-w)/2})$. □

Remark 3.2. The induction needs the short minterm only where no other bound is available. Let $\mathcal{B} \subseteq \mathcal{C}$ be classes of monotone functions closed under restrictions, suppose $H \leq C_{\mathcal{B}} I$ on $\mathcal{B}$, and suppose every nonconstant $f \in \mathcal{C} \setminus \mathcal{B}$ has a minterm or a maxterm with at most w elements. Then $H \leq \max(c_w, C_{\mathcal{B}})\, I$ on $\mathcal{C}$. For example, let $\mathcal{B}$ be the monotone regular functions, which include the monotone threshold functions and satisfy $H \leq 15.1\, I$ [14], and let $\mathcal{C}$ consist of the functions $g \vee h$ with $g \in \mathcal{B}$ and h a monotone DNF of width at most w, on the same variables or not. A restriction of $g \vee h$ is $g_\rho \vee h_\rho$; if $h_\rho = 0$ it lies in $\mathcal{B}$, and otherwise it is constant or has a term of h_ρ, of size at most w, as an implicant. Hence, for instance, $\mathrm{Maj}_n(x) \vee \bigvee_{uw \in E(G)} (x_u \wedge x_w)$ satisfies $H \leq 15.1\, I$ for every graph G on $[n]$.

Proof of Theorem 1.6. Let g be a nonconstant restriction of f, let $L = L(f)$, and let T be a decision tree for g whose expected depth on a uniformly random input is at most L. If δ_i is the probability that T reads the coordinate i, then $\sum_i \delta_i$ is the expected depth, and the theorem of [10] gives

$$\mathrm{Var}(G) \leq \sum_i \delta_i \mathrm{Inf}_i(g) \leq L \max_i \mathrm{Inf}_i(g),$$

where $G = 1 - 2g$. As $\mathrm{Var}(G) \geq 2m(g)$, some coordinate has $4m(g)/\mathrm{Inf}_v(g) - 2 \leq 2L - 2$, and Theorem 1.3 gives $\Delta_v(g) \leq (4.6355 + \Psi(2L-2))\mathrm{Inf}_v(g)$. The restrictions of f form a class closed under restrictions, so the induction of the introduction applies. For the second bound, $\Psi(2L-2) = \log_2(2e) + \log_2 L + \log_2(1 + \sqrt{2L-2}/L)$ and $\sqrt{2L-2}/L \leq 1/\sqrt{2}$ for $L \geq 1$, so $4.6355 + \Psi(2L-2) \leq \log_2 L + 4.6355 + 2.4427 + 0.7716$. □

Proof of Corollary 1.7. (a) Evaluate the terms one after another, reading the unread variables of the current term until one of them is 0 or all of them are 1. For each term at most two new variables are read in expectation, so this tree has expected depth at most $2m$. A restriction of f is a DNF formula with at most m terms, so $L(f) \le 2m$, and $\log_2(2m) + 7.85 = \log_2 m + 8.85$. CNF formulas are handled in the same way, reading variables until one is 1. (b) The least depth of a decision tree for a Boolean function g is at most $C_0(g)C_1(g)$ [1], where $C_b(g)$ is the largest, over the inputs x with $g(x) = b$, of the least number of coordinates whose values at x force the value b. For monotone g, $C_1(g)$ and $C_0(g)$ are the largest sizes of a minterm and of a maxterm, and neither increases under restriction. Hence $L(f) \le w_0 w_1$. (c) For monotone g the certificate complexity $\max(C_0(g), C_1(g))$ equals the sensitivity $s(g)$ [1], and the sensitivity does not increase under restriction. Hence $L(f) \le s(f)^2$. (d) For monotone g the sensitivity equals the block sensitivity, which is at most $2\deg(g)^2$ [1], and the degree does not increase under restriction. Hence $L(f) \le 4k^4$, and $\log_2(4k^4) + 7.85 = 4\log_2 k + 9.85$. □

4. Relative influences and the reduction

Proof of Corollary 1.4. Han's telescoping [7, (3.4)] writes $H(f) = \sum_{k=1}^{n} \mathbb{E}\,\Delta_k(f_x)$, where f_x is f with the coordinates after k fixed to x, uniformly random, and Δ_k is its restriction step at k. Each f_x is monotone. By Theorem 1.3, $\Delta_k(f_x) \le 4.6355\mathrm{Inf}_k(f_x) + G\big(\mathrm{Inf}_k(f_x), m(f_x)\big)$ with $G(x,m) = x\,\Psi(4m/x - 2)$. The function G is the perspective of the concave Ψ composed with the affine map $(x,m) \mapsto (x, 4m - 2x)$, hence jointly concave, and it is increasing in m. Jensen's inequality with $\mathbb{E}\mathrm{Inf}_k(f_x) = \mathrm{Inf}_k(f)$ and $\mathbb{E}\,m(f_x) \le m(f)$ gives the first bound. The second follows from $\Psi(r) \le \log_2\big(2e(r+2)\big)$. □

Proof of Corollary 1.5. Let $C = \max(K, 4.6355 + \Psi(4/\tau - 2))$ and argue by induction on the number of variables. A nonconstant monotone f either has a coordinate with $\mathrm{Inf}_v \ge \tau m$, where Theorem 1.3 gives $\Delta_v \le C\mathrm{Inf}_v$, or satisfies the hypothesis; in both cases restricting v and using the induction hypothesis on f_0, f_1 gives $H(f) \le C\,I(f)$. □

Running the same descent on a hypothetical counterexample shows where it would have to live: if $H(f) > C\,I(f)$ with $C > 7.08$, then, keeping at each step the restriction with the larger excess $H - CI$, one reaches a restriction g of f with $H(g) - CI(g) \ge H(f) - CI(f)$ and $\mathrm{Inf}_v(g) < 2^{9.08-C} m(g)$ for every coordinate v.

5. The strengthened inequality

For a coordinate v put $a_T = \widehat{F}(T)$ and $d_T = \widehat{F}(T \cup v)$ for $T \subseteq [n] \setminus \{v\}$, let $D_vF = \frac{1}{2}\big(F(x^{v\leftarrow 0}) - F(x^{v\leftarrow 1})\big)$, and let

$$\Delta_v^+ = H^+(f) - \tfrac{1}{2}\big(H^+(f_0) + H^+(f_1)\big), \qquad J(a,d) = \phi(a^2+d^2) - \tfrac{1}{2}\big(\phi((a+d)^2) + \phi((a-d)^2)\big) \ge 0.$$

Lemma 5.1. *For every Boolean function and every coordinate v,*

$$\Delta_v^+ = \phi(d_\emptyset^2) + \sum_{T \ne \emptyset} \big(\mathcal{E}(a_T^2, d_T^2) + J(a_T, d_T)\big),$$

$$\big(I - \mathrm{Var}\big)(f) = \mathrm{Var}(D_vF) + \tfrac{1}{2}\big((I - \mathrm{Var})(f_0) + (I - \mathrm{Var})(f_1)\big).$$

Proof. The nonempty sets are the $T \ne \emptyset$ with $v \notin T$ and the sets $T \cup v$, including $\{v\}$; the restriction F_b has the coefficients $a_T \pm d_T$, and $\mathcal{E}(a^2,d^2) + J(a,d) = \phi(a^2) + \phi(d^2) - \frac{1}{2}(\phi((a+d)^2) + \phi((a-d)^2))$. For the second identity, $I(f) = \mathrm{Inf}_v + \frac{1}{2}(I(f_0) + I(f_1))$, the means of F_0, F_1 are $a_\emptyset \pm d_\emptyset$, so that $\mathrm{Var}(f) = \frac{1}{2}(\mathrm{Var}(f_0) + \mathrm{Var}(f_1)) + d_\emptyset^2$, and $\mathrm{Inf}_v - d_\emptyset^2 = \mathbb{E}(D_vF)^2 - (\mathbb{E}D_vF)^2$. □

Consequently, a class closed under restrictions in which every nonconstant member has a coordinate with $\Delta_v^+ \le C\mathrm{Var}(D_vF)$ satisfies (1), by induction on the number of variables.

Lemma 5.2. *For monotone f and every v with $\mathrm{Inf}_v > 0$, $\Delta_v^+ \le \mathrm{Var}(D_vF)\big(4/\ln 2 + \log_2(e(1+1/\mathrm{Inf}_v))\big)$. In particular $\Delta_v^+ \le 8.80\mathrm{Var}(D_vF)$ when $\mathrm{Inf}_v \ge \frac{1}{2}$.*

Proof. For monotone f, D_vF takes the values 0 and 1, so $d_\emptyset = \mathrm{Inf}_v$ and $\sum_{T\ne\emptyset} d_T^2 = \mathrm{Var}(D_vF) = \mathrm{Inf}_v(1-\mathrm{Inf}_v)$. The function $A = \frac{1}{2}(F_0+F_1)$ has $\widehat{A}(T) = a_T$ and vanishes exactly where v is pivotal, so $\sum_{T\ne\emptyset} a_T^2 \le \mathbb{E}A^2 = 1-\mathrm{Inf}_v$. First, $\phi(\mathrm{Inf}_v^2) = \frac{2}{\ln 2}\mathrm{Inf}_v^2 \ln\frac{1}{\mathrm{Inf}_v} \le \frac{2}{\ln 2}\mathrm{Var}(D_vF)$. Next let $M = \sum_{T\ne\emptyset}\min(a_T^2, d_T^2) \le \mathrm{Var}(D_vF)$ and $S = \sum_{T\ne\emptyset}(a_T^2+d_T^2)$. By the pointwise bound in the proof of Lemma 2.1, $\sum_{T\ne\emptyset} J(a_T,d_T) \le \frac{2}{\ln 2}M$. Since $\mathcal{E}(X,Y) \le \min(X,Y)\log_2\big(e(X+Y)/\min(X,Y)\big)$, the log-sum inequality gives

$$\sum_{T\ne\emptyset}\mathcal{E}(a_T^2,d_T^2) \le M\log_2(eS/M),$$

and as $x\log_2(eS/x)$ increases for $x \le S$, this is at most $\mathrm{Var}(D_vF)\log_2(eS/\mathrm{Var}(D_vF))$, where $S/\mathrm{Var}(D_vF) \le 1 + (1-\mathrm{Inf}_v)/\mathrm{Var}(D_vF) = 1 + 1/\mathrm{Inf}_v$. For $\mathrm{Inf}_v \ge \frac{1}{2}$ the constant is $4/\ln 2 + \log_2(3e) < 8.80$. □

Proof of Theorem 1.8. Let $f \in \mathcal{C}$ be nonconstant and let v be the coordinate of Lemma 3.1, so that $4m/\mathrm{Inf}_v - 2 \le 2^{w+1}-4$; for the last statement, let f be a nonconstant restriction of the given function and v the coordinate found in the proof of Theorem 1.6, with $4m/\mathrm{Inf}_v - 2 \le 2L-2$, and replace $2^{w+1}-4$ by $2L-2$ below. If $\mathrm{Inf}_v \ge \frac{1}{2}$, Lemma 5.2 gives $\Delta_v^+ \le 8.80\mathrm{Var}(D_vF)$. Otherwise Lemma 5.1 gives $\Delta_v^+ \le \phi(\mathrm{Inf}_v^2) + \sum_{T\ne\emptyset}\mathcal{E}(a_T^2,d_T^2) + \mathrm{JS}_v$. The middle sum is the part of B_v bounded in the proof of Lemma 2.3 by $\mathrm{Inf}_v\Psi(4m/\mathrm{Inf}_v - 2)$, and $\mathrm{JS}_v \le \frac{2}{\ln 2}\mathrm{Inf}_v$ by Lemma 2.1. As $\phi(x^2) \le \frac{2}{e\ln 2}\,x < 1.0615\,x$ and $\mathrm{Var}(D_vF) = \mathrm{Inf}_v(1-\mathrm{Inf}_v) \ge \frac{1}{2}\mathrm{Inf}_v$, we get $\Delta_v^+ \le C_w^+\mathrm{Var}(D_vF)$, and Lemma 5.1 completes the induction. The estimate $C_w^+ \le 2w + 14.8$ follows as in the proof of Theorem 1.1, and the estimate $2\log_2 L + 14.33$ as in the proof of Theorem 1.6, since $2(1.0615 + 2/\ln 2 + 2.4427 + 0.7716) < 14.33$. □

Proof of Corollary 1.9. By [11, Theorem 1], if $g_1,\dots,g_k$ on disjoint sets of variables satisfy (1) with constant C and G satisfies the biased form of (1) with constant C under the product measure with the means of the g_j, then $G(g_1,\dots,g_k)$ satisfies (1) with constant C. The leaves satisfy it with C_w^+ by Theorem 1.8, AND, OR, XOR and threshold gates with $2+4\log_2 e < C_w^+$ or 760 [14], and a monotone DNF or CNF gate of width at most w with balanced inputs needs the inequality only under the uniform measure, where Theorem 1.8 gives C_w^+. Induction on the depth of F and [11, Proposition 2.1] give the claim. □

6. Product measures

Let μ be a product measure on $\{0,1\}^n$ with $q_i = \mu(x_i=1) \in (0,1)$, put $\sigma_i^2 = 4q_i(1-q_i)$ and $\sigma_A^2 = \prod_{i\in A}\sigma_i^2$, and let $\chi_A^\mu = \prod_{i\in A}(x_i-q_i)/\sqrt{q_i(1-q_i)}$ be the orthonormal basis of $L^2(\mu)$. For $F = 1-2f = \sum_A \widetilde{F}(A)\chi_A^\mu$ put $P_\mu(A) = \widetilde{F}(A)^2$, $I_\mu(f) = \sum_A |A|P_\mu(A)$, $\mathrm{Var}_\mu(f) = 1-P_\mu(\emptyset)$, and

$$L_\mu(f) = \sum_{A\ne\emptyset} P_\mu(A)\log_2\frac{\sigma_A^2}{P_\mu(A)}, \qquad R_\mu(f) = I_\mu(f) - \mathrm{Var}_\mu(f),$$

so that the biased form of (1) reads $L_\mu(f) \le C\,R_\mu(f)$. Fix a coordinate v and write $q = q_v$, $\sigma^2 = \sigma_v^2$, $\tau = \sqrt{q/(1-q)}$, and $a_T = \widetilde{F}(T)$, $d_T = \widetilde{F}(T\cup v)$ for $T \not\ni v$. The restrictions f_0, f_1 to $x_v = 0,1$, with μ on the other coordinates, have the coefficients $a_T - \tau d_T$ and $a_T + d_T/\tau$; they are weighted with $1-q$ and q, and the mixture of their spectra is $a_T^2 + d_T^2$ because the cross

terms cancel. Put $\mathrm{Inf}_v^\mu = \sum_T d_T^2$ and

$$J_q(a,d) = \phi(a^2+d^2) - (1-q)\,\phi\big((a-\tau d)^2\big) - q\,\phi\big((a+d/\tau)^2\big) \ge 0.$$

Lemma 6.1. *For every Boolean function and every coordinate v,*

$$L_\mu(f) - (1-q)L_\mu(f_0) - q\,L_\mu(f_1) = d_\emptyset^2 \log_2 \frac{\sigma^2}{d_\emptyset^2} + \sum_{T\neq\emptyset}\big(\mathcal{E}(a_T^2,d_T^2) + J_q(a_T,d_T)\big) - \log_2\frac{1}{\sigma^2}\sum_{T\neq\emptyset} d_T^2,$$

$$R_\mu(f) - (1-q)R_\mu(f_0) - q\,R_\mu(f_1) = \sum_{T\neq\emptyset} d_T^2.$$

Proof. The factors σ_A^2 with $A \not\ni v$ are the same in f and in its restrictions, and each set $T\cup v$ carries the extra factor σ^2; with the mixture $a_T^2+d_T^2$ this gives the first identity, as in Lemma 5.1. For the second, I_μ drops by Inf_v^μ and Var_μ by $d_\emptyset^2$. □

The Jensen–Shannon part is bounded by the logarithmic Sobolev constant of the biased two-point space, and the factor σ^2 pays for it.

Lemma 6.2. *In this notation $J_q(a,d) \le \frac{c(q)}{\ln 2}\,d^2$ for all real a,d, where $c(q) = \ln((1-q)/q)/(1-2q)$ and $c(\frac12) = 2$. The constant $c(q)$ is best possible, and $c(q) \le 2 + \ln(1/\sigma^2)$.*

Proof. Put $u = a - \tau d$ and $w = a + d/\tau$. Then $(1-q)u^2 + qw^2 = a^2+d^2$, so $J_q(a,d)$ is $1/\ln 2$ times $\mathrm{Ent}_\nu(g^2)$ for the function g with values $|u|,|w|$ on the two-point space with measure $\nu = (1-q,q)$, and $d^2 = q(1-q)(w-u)^2 \ge \mathrm{Var}_\nu(g)$. The logarithmic Sobolev inequality of this space with the optimal constant of Diaconis and Saloff-Coste [3, Theorem A.1] gives $\mathrm{Ent}_\nu(g^2) \le c(q)\mathrm{Var}_\nu(g)$. When u and w have the same sign, $d^2 = \mathrm{Var}_\nu(g)$, so the optimality of $c(q)$ in that inequality shows that it cannot be lowered here. For the last claim put $x = |1-2q| < 1$. Then $c(q) = \frac1x \ln\frac{1+x}{1-x} = 2\sum_{k\ge0}\frac{x^{2k}}{2k+1}$ and $2 + \ln\frac{1}{\sigma^2} = 2 + \ln\frac{1}{1-x^2} = 2 + \sum_{k\ge1}\frac{x^{2k}}{k}$, and $\frac{2}{2k+1} \le \frac1k$ for $k \ge 1$. □

For monotone f let $D = \{f=0\}$, let μ_D be μ conditioned on D, and let μ' be the marginal of μ on the coordinates other than v. Call y blocked or free as in Lemma 2.2; the blocked y are the points at which v is pivotal. Let $\pi = \mu'(v \text{ pivotal})$ and $\theta = \mu_D(y \text{ blocked})$.

Lemma 6.3. *For monotone f and $\pi > 0$, $d_\emptyset^2 = \sigma^2\pi^2$, $\mathrm{Inf}_v^\mu = \sigma^2\pi$, and*

$$\sum_{T\neq\emptyset}\mathcal{E}(a_T^2,d_T^2) \le \mathrm{Inf}_v^\mu \log_2\Big(e\Big(1 + \frac{2(1-q)}{q} + \frac{2(1-\theta)}{q\theta}\Big)\Big).$$

Proof. Let B and Fr be the sets of blocked and free y. Under μ_D the coordinate x_v is 0 over a blocked y and has the law $(1-q,q)$ over a free one, so with $\alpha(T) = \mathbb{E}_{\mu_D}[\chi_T^\mu \mathbf{1}_{\mathrm{B}}]$ and $\gamma(T) = \mathbb{E}_{\mu_D}[\chi_T^\mu \mathbf{1}_{\mathrm{Fr}}]$ we get $\mathbb{E}_{\mu_D}\chi^\mu_{T\cup v} = -\tau\alpha(T)$ and $\mathbb{E}_{\mu_D}\chi_T^\mu = \alpha(T)+\gamma(T)$. As $F = 2\cdot\mathbf{1}_D - 1$, $d_T = -2\mu(D)\tau\alpha(T)$ for every T and $a_T = 2\mu(D)(\alpha(T)+\gamma(T))$ for $T \neq \emptyset$. The marginal of μ_D on y has density $(1-q)/\mu(D)$ on B and $1/\mu(D)$ on Fr with respect to μ', so Parseval for μ' gives $\sum_T \alpha(T)^2 = (1-q)^2\pi/\mu(D)^2$ and $\sum_T \gamma(T)^2 = \mu'(\mathrm{Fr})/\mu(D)^2$. Hence $\mathrm{Inf}_v^\mu = 4q(1-q)\pi$, and $\alpha(\emptyset) = (1-q)\pi/\mu(D)$ gives $d_\emptyset^2 = \sigma^2\pi^2$. For $T \neq \emptyset$ with $\alpha(T) \neq 0$ put $s_T = \gamma(T)/\alpha(T)$. By $\mathcal{E}(X,Y) \le Y\log_2(e(X+Y)/Y)$ and $(1+s)^2 \le 2+2s^2$, the pair contributes at most $w_T g(s_T^2)$, where $w_T = 4\mu(D)^2\tau^2\alpha(T)^2$ and $g(z) = \log_2\big(e(1+2/\tau^2+2z/\tau^2)\big)$ is concave and increasing. The weights have total $W \le \mathrm{Inf}_v^\mu$ and $\sum_T w_T s_T^2 \le 4\tau^2\mu'(\mathrm{Fr})$, so Jensen's inequality and the monotonicity of $W \mapsto Wg(c/W)$, as in the proof of Lemma 2.3, bound the sum by $\mathrm{Inf}_v^\mu\, g(4\tau^2\mu'(\mathrm{Fr})/\mathrm{Inf}_v^\mu)$. Finally $\mu(D) = (1-q)\pi + \mu'(\mathrm{Fr})$ and $\theta = (1-q)\pi/\mu(D)$ give $8\mu'(\mathrm{Fr})/\mathrm{Inf}_v^\mu = 2(1-\theta)/(q\theta)$. □

Lemma 6.4. *Let f be monotone with a minterm M of size k, and suppose $q_i \ge \eta$ for $i \in M$, where $0 < \eta < 1$. Then some $v \in M$ has $\theta \ge \eta^{k-1}(1-\eta)$.*

Proof. For $k = 1$ every y with $(y, 0) \in D$ is blocked, so $\theta = 1$. Let $k \ge 2$, $M = \{v_1, \dots, v_k\}$, and let δ be the largest of the values of θ at $v_1, \dots, v_k$. Under μ_D the conditional probability that $x_{v_j} = 1$ given the other coordinates is $q_{v_j} \ge \eta$ over a free point and 0 over a blocked one, so $p_j = \mu_D(x_{v_2} = \cdots = x_{v_j} = 1)$ satisfies $p_1 = 1$ and $p_{j+1} \ge \eta(p_j - \delta)$, whence $p_k \ge \eta^{k-1} - \delta\eta/(1-\eta)$. When $x_{v_2} = \cdots = x_{v_k} = 1$ the coordinate v_1 is blocked, so $\delta \ge p_k$, and therefore $\delta \ge \eta^{k-1}(1-\eta)$. □

Proof of Theorem 1.10. The dual $f^*(x) = 1 - f(\mathbf{1} - x)$, under the measure with the biases $1 - q_i$, has the same P_μ, σ_A^2, I_μ and Var_μ, and its restrictions correspond to those of f; so we may assume that the nonconstant $f \in \mathcal{C}$ has a minterm of size $k \le w$, and we restrict the coordinate v of Lemma 6.4, for which $\theta \ge \eta^{k-1}(1-\eta) \ge \eta^{k-1}/2$. Write $\rho = \sum_{T \ne \emptyset} d_T^2 = \sigma^2 \pi(1-\pi)$. By Lemma 6.2, $\sum_{T \ne \emptyset} J_q(a_T, d_T) - \log_2(1/\sigma^2)\rho \le \frac{2}{\ln 2}\rho$, and $d_\emptyset^2 \log_2(\sigma^2/d_\emptyset^2) = 2\mathrm{Inf}_v^\mu \, \pi \log_2(1/\pi)$ by Lemma 6.3.

If $\pi \le \frac{1}{2}$, then $\mathrm{Inf}_v^\mu \le 2\rho$ and $2\pi \log_2(1/\pi) \le \frac{2}{e \ln 2} < 1.0615$. In Lemma 6.3, $\frac{2(1-q)}{q} + \frac{2(1-\theta)}{q\theta} \le \frac{4}{\eta\theta} \le 8\eta^{-k}$, so the erasure part is at most $\mathrm{Inf}_v^\mu(\log_2(9e) + k\log_2(1/\eta))$. With Lemma 6.1 the step of L_μ is at most $\rho\big(2 \cdot 1.0615 + 2\log_2(9e) + 2/\ln 2 + 2k\log_2(1/\eta)\big) < (14.24 + 2k\log_2(1/\eta))\,\rho$.

If $\pi > \frac{1}{2}$, then $2\mathrm{Inf}_v^\mu \pi \log_2(1/\pi) \le \frac{2}{\ln 2}\mathrm{Inf}_v^\mu(1-\pi) = \frac{2}{\ln 2}\rho$. For the erasure part argue as in Lemma 5.2: $\sum_{T \ne \emptyset} \mathcal{E}(a_T^2, d_T^2) \le \rho \log_2(eS/\rho)$ with $S = \sum_{T \ne \emptyset}(a_T^2 + d_T^2)$. The function $(1-q)F_0 + qF_1$ has the coefficients a_T, equals $1 - 2q$ where v is pivotal and ± 1 elsewhere, so its variance is at most $4(1-\pi)$; hence $S/\rho \le 1 + 4/(\sigma^2\pi) \le 9/\sigma^2$, and $\sigma^2 \ge 4\eta(1-\eta) \ge 2\eta$. The step of L_μ is then at most $\rho\big(4/\ln 2 + \log_2(9e/2) + \log_2(1/\eta)\big) < (9.39 + \log_2(1/\eta))\,\rho$.

In both cases Lemma 6.1 bounds the step of L_μ by $(14.24 + 2w\log_2(1/\eta))$ times the step of R_μ, the restrictions of f lie in $\mathcal{C}$, and the measure on their coordinates has biases in $[\eta, 1-\eta]$; induction on the number of variables completes the proof. □

7. Remarks

Remark 7.1. The logarithm in Theorem 1.3 is needed. For odd $n = 2m' + 1$ and y^* with m' ones, the monotone function $g_n(y, z) = \mathrm{Maj}_n(y) \vee (z \wedge [y = y^*])$ has $\mathbb{P}[g_n = 0] = \frac{1}{2} - 2^{-n-1}$ and $\mathrm{Inf}_z = 2^{-n}$, so $\log_2(4m/\mathrm{Inf}_z) < n + 1$. For $T \subseteq [n]$ the pair $(T, T \cup z)$ has weights u_T^2 and v_T^2 with $v_T = \pm 2^{-n}\chi_T(y^*)$ and $u_T = \widehat{M}(T) - 2^{-n}\chi_T(y^*)$, where $M = 1 - 2\,\mathrm{Maj}_n$. For $|T| = 2j+1$, $|\widehat{M}(T)| = 2^{1-n}\binom{2m'}{m'}\binom{m'}{j}/\binom{2m'}{2j}$, and log-concavity gives $\binom{2m'}{2j} \le (m'+1)\binom{m'}{j}^2$; hence $|\widehat{M}(T)| \ge 2^{-n}2^{m'}/(\sqrt{m'}(m'+1))$, which for $n \ge 11$ is at least 2^{1-n}, so that $|u_T| \ge \frac{1}{2}|\widehat{M}(T)|$ and $4^n u_T^2 \ge 2^n/((n-1)(n+1)^2)$. Using $\mathcal{E}(X, Y) \ge Y \log_2(1 + X/Y)$ on the 2^{n-1} sets of odd size gives $B_z \ge 2^{n-1}4^{-n}\log_2\big(2^n/((n-1)(n+1)^2)\big) \ge \frac{1}{2}(n - 3\log_2(n+1))\mathrm{Inf}_z$. The exact ratios B_z/Inf_z for $n = 3, 5, \dots, 17$ are $3.52, 4.24, 4.99, 5.78, 6.62, 7.48, 8.37, 9.27$.

Numerically the coefficient 1 of the logarithm is attained as well. Majority loses a factor 2 because $\widehat{M}$ vanishes on the sets of even size, where the pairs cost nothing. Replace Maj_n by the threshold function $[\,|y| \ge k\,]$, so that $M = 1 - 2[\,|y| \ge k\,]$, and y^* by a point with $k-1$ ones. Then $2^n\widehat{M}(T) = 2K_{k-1}(|T| - 1; n-1)$ for $T \ne \emptyset$, where $K_j(x; N) = \sum_i (-1)^i\binom{x}{i}\binom{N-x}{j-i}$ is the Krawtchouk polynomial, and the pairs can be summed exactly in terms of $|T|$ and $|T \cap \mathrm{supp}\, y^*|$. For the best k and $n = 21, 31, 41, 61, 81$ the ratio Δ_z/Inf_z is $20.64, 29.85, 39.39, 58.79, 78.41$, against $\log_2(4m/\mathrm{Inf}_z) = 20.62, 30.17, 41.09, 60.29, 80.58$; the bound of Theorem 1.3 exceeds it by 6.1 to 8.3.

Proposition 7.2. *For every K there is a monotone function f with a unique coordinate z of largest influence and $\Delta_z(f) > K \operatorname{Inf}_z(f)$. For instance, if every y_j in the function g_5 of Remark 7.1 is replaced by the majority of 109 new variables, the resulting monotone function has 546 variables, z is its unique coordinate of largest influence, and $\Delta_z/\operatorname{Inf}_z = 6.6596$.*

Proof. Let g be a Boolean function of $y_1, \dots, y_r$ and z, let $h_1, \dots, h_r$ be balanced Boolean functions on pairwise disjoint sets of new variables, and let $f = g(h_1, \dots, h_r, z)$. With $G = 1 - 2g$ and $H_j = 1 - 2h_j$, the function $F = 1 - 2f$ is obtained from the Fourier expansion of G by substituting H_j for the character of y_j, and $\widehat{H_j}(\emptyset) = 0$. Hence $\widehat{F}\big(Z \cup \bigcup_{j\in A} U_j\big) = \widehat{G}(A \cup Z) \prod_{j\in A} \widehat{H_j}(U_j)$ for $A \subseteq [r]$, $Z \subseteq \{z\}$ and nonempty sets U_j of variables of h_j, and all other coefficients vanish: the spectral sample of f is that of g with each y_j in it replaced by an independent spectral sample of h_j. Therefore $\operatorname{Inf}_z(f) = \operatorname{Inf}_z(g)$, a variable u of h_j has influence $\operatorname{Inf}_{y_j}(g)\operatorname{Inf}_u(h_j)$, and $H(f) = H(g) + \sum_j \operatorname{Inf}_{y_j}(g) H(h_j)$. The restrictions of f at z are $g_b(h_1, \dots, h_r)$, so the same formula holds for them, and since $\operatorname{Inf}_{y_j}(g) = \frac{1}{2}(\operatorname{Inf}_{y_j}(g_0) + \operatorname{Inf}_{y_j}(g_1))$ the terms with $H(h_j)$ cancel in Δ_z: $\Delta_z(f) = \Delta_z(g)$.

Take $g = g_n$ from Remark 7.1 and $h_j = \operatorname{Maj}_k$ with k odd; then f is monotone. As $k \to \infty$ the influences of Maj_k tend to 0, while $\operatorname{Inf}_z(f) = 2^{-n}$, so for large k the coordinate z is the unique coordinate of largest influence, and $\Delta_z(f)/\operatorname{Inf}_z(f) = \Delta_z(g_n)/\operatorname{Inf}_z(g_n) \geq B_z(g_n)/\operatorname{Inf}_z(g_n) \geq \frac{1}{2}(n - 3\log_2(n+1))$ for $n \geq 11$. For the instance, the influences of g_5 at the y_j are at most $13/32$, those of Maj_{109} equal $\binom{108}{54}2^{-108} < 0.0766$, and $\frac{13}{32} \cdot 0.0766 < 2^{-5} = \operatorname{Inf}_z$; the ratio is that of g_5, computed exactly. □

With the threshold functions of Remark 7.1 in place of g_n the same construction gives monotone functions whose unique coordinate v of largest influence has $\Delta_v/\operatorname{Inf}_v$ within a few units of $\log_2(4m/\operatorname{Inf}_v)$ in the computed range, so numerically the logarithm in Theorem 1.3 is attained with its coefficient even at such a coordinate.

Remark 7.3. When every influence is small compared with m, ε is small at every coordinate, the free fibres carry almost all of μ, and Lemma 2.3 is no better than the bound $H(f) \leq \sum_v h(\operatorname{Inf}_v)$. The argument used only the second moments of a and b. What can happen beyond them is shown by majority on an odd number of variables: $\widehat{F}$ vanishes on sets of even size, so $a(T) = 0$ when $|T|$ is odd and $a(T) + b(T) = 0$ when $|T|$ is even and nonempty. Every pair then has one weight equal to zero and $B_v = 0$, although $\varepsilon \sim n^{-1/2}$. A proof of the conjecture for monotone functions along these lines would need an estimate of this alignment between the blocked and the free parts of the down-set.

Numerically the step bound holds at every coordinate of largest influence: for every monotone function of at most six variables, and for perfect matchings, tribes, majority and the property of containing a triangle, every coordinate v of largest influence has $\Delta_v \leq 6.01 \operatorname{Inf}_v$, the largest ratio being that of a perfect matching of eight edges; over each of these classes the largest ratio at such coordinates equals the largest value of $\min_v \Delta_v/\operatorname{Inf}_v$. For a perfect matching of k edges the spectrum is a product over the edges, and the ratio, computed exactly, increases to 6.0037 at $k = 8$ and then decreases to 5.735 as $k \to \infty$. Among disjunctions of $k \leq 40$ disjoint copies of any monotone function of at most four variables, which have the same product structure, the largest ratio is again this one, and a search by simulated annealing over monotone functions of eight to twelve variables found nothing above 5.72. These searches do not reach the functions of Proposition 7.2, which need hundreds of variables: in general the coordinate of largest influence does not work, and a proof has to choose the coordinate by more than the size of its influence. The averaged form in Proposition 7.4 below needs no choice. On every function computed $\sum_v B_v \leq 3.61\, I$; the largest value over the monotone functions of six variables is 3.03, and 3.61 is again the perfect matching of eight edges.

The erasure terms carry the whole conjecture, and in a form that involves one step of the spectral sample at a time. Write $\log_2^+ t = \max(\log_2 t, 0)$ and

$$R(f) = \sum_{S:\, P(S)>0} P(S) \sum_{v \in S} \log_2^+ \frac{P(S \setminus \{v\})}{P(S)}.$$

Proposition 7.4. *For every Boolean function f,*

$$R(f) \le \sum_v B_v \le \min\bigl(H(f),\, R(f) + \log_2(2e)\, I(f)\bigr).$$

Consequently, if $\mathcal{C}$ is a class of Boolean functions closed under restrictions, then $H \le CI$ on $\mathcal{C}$ implies $R \le \sum_v B_v \le CI$ on $\mathcal{C}$, and $R \le KI$ on $\mathcal{C}$ implies $H \le (K + \log_2(2e) + 2/\ln 2)\, I$ on $\mathcal{C}$.

Proof. By the chain rule, $H(f) = \sum_v H(\mathbf{1}[v \in S] \mid \mathbf{1}[u \in S],\, u < v)$, and each term is at least $H(\mathbf{1}[v \in S] \mid S \setminus \{v\}) = B_v$. For $X \ge 0$ and $Y > 0$,

$$Y \log_2^+(X/Y) \le Y \log_2 \frac{X+Y}{Y} \le \mathcal{E}(X,Y) \le Y \log_2 \frac{e(X+Y)}{Y} \le Y\bigl(\log_2(2e) + \log_2^+(X/Y)\bigr),$$

and $\mathcal{E}(X,0) = 0$. Applying this to $X = P(T)$, $Y = P(T \cup v)$ and summing over v and $T \not\ni v$, that is over the pairs (S, v) with $v \in S = T \cup v$, gives the two bounds on $\sum_v B_v$. If $R \le KI$ on $\mathcal{C}$, then every nonconstant $f \in \mathcal{C}$ has $\sum_v B_v \le (K + \log_2(2e)) \sum_v \mathrm{Inf}_v$, and $B_v = 0$ when $\mathrm{Inf}_v = 0$, so some v with $\mathrm{Inf}_v > 0$ has $B_v \le (K + \log_2(2e))\mathrm{Inf}_v$; with Lemma 2.1 this gives $\Delta_v \le (K + \log_2(2e) + 2/\ln 2)\mathrm{Inf}_v$, and the induction of the introduction applies. □

With $\mathcal{C}$ the class of all Boolean functions, the conjecture is thus equivalent to $R \le KI$: removing one element of the spectral sample should increase its weight by at most a bounded factor, on average in the logarithm. For monotone f and $S = T \cup \{v\}$ with $T \ne \emptyset$ the ratio is $P(T)/P(S) = \bigl(1 + b(T)/a(T)\bigr)^2$, and Lemma 2.3 bounds its logarithmic average using only the second moments of a and b.

Remark 7.5. The constant of Theorem 1.10 grows like $w \log_2(1/\eta)$, and the argument gives nothing as the biases tend to 0 or 1: Lemma 6.3 uses only second moments, and under a product measure with small biases every coordinate of a large matching has small relative influence. The restriction step is not what fails there. For the disjunction of k disjoint conjunctions of two inputs, all of bias q, with $k \approx \ln 2/q^2$ so that the output is balanced, the biased spectral sample is a product over the pairs and can be computed exactly: at every input v the conditional entropy $H(\mathbf{1}[v \in S] \mid S \setminus \{v\})$ is $o(\mathrm{Inf}_v^\mu)$ as $q \to 0$, where $\mathrm{Inf}_v^\mu = \sum_{A \ni v} P_\mu(A)$. The loss is in the second-moment estimate of Lemma 6.3, which does not see that each pair of weights $(P_\mu(T), P_\mu(T \cup v))$ is lopsided. Numerically the induction on coordinates itself goes through: with L_μ and R_μ as in Section 6, every monotone 2-DNF gate tested (fourteen graphs on up to six inputs, 261 bias vectors each) has a coordinate at which $L_\mu(G) - \mathbb{E}_b L_\mu(G_b) \le (2 + 4\log_2 e)\bigl(R_\mu(G) - \mathbb{E}_b R_\mu(G_b)\bigr)$, the restriction G_b being taken with probability $\mu(y_v = b)$; the extreme case is a single edge with both biases near 1, and on random 2- and 3-DNF gates with up to eight inputs, with a local search on the biases, the largest ratio is 6.87. By Lemmas 6.1 and 6.2, what is missing for a constant independent of the biases is a bound on the erasure part $\sum_T \mathcal{E}(a_T^2, d_T^2)$ that sees the lopsided pairs. For a monotone 2-DNF the uniform measure on the zeros, taken under a product measure with biases q_v, is the hard-core model with activities $q_v/(1-q_v)$. Numerically the biased form holds with the constant $2 + 4\log_2 e$ of AND and OR. Over all monotone functions of at most five variables, and over random monotone 2- and 3-DNFs with up to ten variables, with the biases chosen by local search, no ratio of the two sides exceeds it; values above it computed in floating point at biases near 10^{-7} fall below it in 60-digit arithmetic.

Remark 7.6. The statements were checked numerically. Lemma 2.2 holds to within 10^{-15}, and Theorem 1.3 and Corollary 1.4 hold, for all 7,828,354 monotone functions of six variables. On monotone DNF and CNF formulas of widths two to four with up to 20 variables and 120 terms (paths, cycles, grids, hypercubes, complete, complete bipartite, crown, Paley, random regular and multipartite graphs, and random formulas), the largest ratio of Δ_v to the bound of Theorem 1.3 was 0.68, and the lower bound of Lemma 3.1 held in every case; a perfect matching attains $\varepsilon = \frac{1}{6}$. The largest value of H/I found for graph formulas was 4.52, for a perfect matching of ten edges. For the strengthened inequality, the identity of Lemma 5.1 holds to within 10^{-14} for all monotone functions of six variables, the two bounds used in the proof of Theorem 1.8 are attained to at most 0.75 and 0.79 of their values, and $H^+/(I - \mathrm{Var})$ stays below $0.47\,C_w^+$ for DNFs of width w. On the formula families above $H^+/(I - \mathrm{Var}) \le 6.3$, and on conjunctions, disjunctions and majorities of two to four monotone 2-DNFs with up to 20 variables $H/I \le 4.27$. For Theorem 1.6, the least expected depth of a decision tree was computed by dynamic programming over subcubes for all monotone functions of at most five variables and for random monotone DNFs with eight to ten variables: the influence bound of [10] used in the proof held in every case, with equality for 87 of the 7,579 nonconstant functions of five variables, and H/I stayed below 0.44 times the bound of Theorem 1.6. On 400 random monotone DNF formulas with up to 10 variables and 29 terms, $L(f) \le 2m$ held with largest ratio 0.992. The quantity R of Proposition 7.4 is small in every case computed: over all monotone functions of six variables the largest value of R/I is 1.34, against 3.74 for H/I; majority has $R = 0$; and over the families above the largest value is 1.85, for a perfect matching of eight edges. The scalar inequalities in the proof of Lemma 2.3 (concavity and monotonicity of Ψ, monotonicity of $W \mapsto W\Psi(c/W)$, and the resulting Jensen bound) and the arithmetic in the proof of Lemma 3.1 were checked in Lean 4 with Mathlib.

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Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies

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Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel
Email address: yitzchak.shmalo@gmail.com